微分几何习题解答*

2020 Autumn

§1 第四章习题解答

1 (1).$g^{\alpha\beta}g_{\alpha\beta}=\delta^{\alpha}_{\alpha}=2$，注意重复指标表示求和，$\alpha,\beta=1,2$；

(2).注意

$$(g_{\alpha\beta})=\begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix} \quad (g^{\alpha\beta})=\frac{1}{g}\begin{pmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{pmatrix}$$

由

$$\begin{aligned} \Gamma^1_{1\alpha}+\Gamma^2_{2\alpha} &= \Gamma^k_{k\alpha} \\ &= \frac{1}{2}g^{kh}(\partial_k g_{\alpha h}+\partial_\alpha g_{kh}-\partial_h g_{k\alpha}) \\ &= \frac{1}{2}g^{kh}\partial_\alpha g_{kh} \\ &= \frac{1}{2}(g^{11}\partial_\alpha g_{11}+g^{22}\partial_\alpha g_{22}+2g^{12}\partial_\alpha g_{12}) \\ &= \frac{1}{2g}(g_{22}\partial_\alpha g_{11}+g_{11}\partial_\alpha g_{22}-2g_{12}\partial_\alpha g_{12}) \\ &= \frac{1}{2g}\partial_\alpha g \end{aligned}$$

而

$$\frac{\partial \ln\sqrt{g}}{\partial u^\alpha}=\frac{1}{2g}\partial_\alpha g$$

所以

$$\frac{\partial \ln\sqrt{g}}{\partial u^\alpha}=\Gamma^1_{1\alpha}+\Gamma^2_{2\alpha}$$

2. (1).

$$\tilde{g}_{ij}=<\frac{\partial r}{\partial \tilde{u}^i},\frac{\partial r}{\partial \tilde{u}^j}>=<\frac{\partial r}{\partial u^\alpha}a^\alpha_i,\frac{\partial r}{\partial u^\beta}a^\beta_j>=g_{\alpha\beta}a^\alpha_i a^\beta_j$$

$$\tilde{b}_{ij}=-<\frac{\partial r}{\partial \tilde{u}^i},\frac{\partial n}{\partial \tilde{u}^j}>=-<\frac{\partial r}{\partial u^\alpha}a^\alpha_i,\frac{\partial n}{\partial u^\beta}a^\beta_j>=b_{\alpha\beta}a^\alpha_i a^\beta_j$$

由$(\tilde{g}_{ij})=(a^\alpha_i)(g_{\alpha\beta})(a^\beta_j)$，两边取逆得,$(\tilde{g}^{ij})=(a^\beta_j)^{-1}(g^{\alpha\beta})(a^\alpha_i)^{-1}$，再左乘$(a^\beta_j)$，右乘$(a^\alpha_i)$，即得

$$(a^\beta_j)(\tilde{g}^{ij})(a^\alpha_i)=(g^{\alpha\beta})$$

即$g^{\alpha\beta}=\tilde{g}^{ij}a^\alpha_i a^\beta_j$；

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(2). 由曲面S的自然标架的运动方程知

$$\mathbf{r}_{\alpha\beta} = \Gamma^{\gamma}_{\alpha\beta}\mathbf{r}_{\gamma} + b_{\alpha\beta}\mathbf{n}$$

同样有

(1.1) $$\tilde{\mathbf{r}}_{ij} = \tilde{\Gamma}^{k}_{ij}\tilde{\mathbf{r}}_{k} + \tilde{b}_{ij}\mathbf{n}$$

(1.1)式左边为

$$\begin{aligned}\tilde{\mathbf{r}}_{ij} &= \mathbf{r}_{\alpha\beta}a^{\alpha}_{i}a^{\beta}_{j} + \mathbf{r}_{\gamma}\frac{\partial a^{\gamma}_{i}}{\partial \tilde{u}^{j}} \\ &= (\Gamma^{\gamma}_{\alpha\beta}a^{\alpha}_{i}a^{\beta}_{j} + \frac{\partial a^{\gamma}_{i}}{\partial \tilde{u}^{j}})\mathbf{r}_{\gamma} + b_{\alpha\beta}a^{\alpha}_{i}a^{\beta}_{j}\mathbf{n} \\ &= (\Gamma^{\gamma}_{\alpha\beta}a^{\alpha}_{i}a^{\beta}_{j} + \frac{\partial a^{\gamma}_{i}}{\partial \tilde{u}^{j}})\mathbf{r}_{\gamma} + \tilde{b}_{ij}\mathbf{n}\end{aligned}$$

(1.1)式右边为

$$\tilde{\Gamma}^{k}_{ij}\tilde{\mathbf{r}}_{k} + \tilde{b}_{ij}\mathbf{n} = \tilde{\Gamma}^{k}_{ij}\mathbf{r}_{\gamma}a^{\gamma}_{k} + \tilde{b}_{ij}\mathbf{n}$$

由于$\{\mathbf{r}_1, \mathbf{r}_2, \mathbf{n}\}$线性无关，故比较系数得

$$\Gamma^{\gamma}_{\alpha\beta}a^{\alpha}_{i}a^{\beta}_{j} + \frac{\partial a^{\gamma}_{i}}{\partial \tilde{u}^{j}} = \tilde{\Gamma}^{k}_{ij}a^{\gamma}_{k}$$

注意矩阵(a^{α}_{i})与$(\tilde{a}^{i}_{\alpha})$互逆，即$a^{\alpha}_{i}\tilde{a}^{j}_{\alpha} = \delta^{j}_{i}$。所以

$$\tilde{\Gamma}^{k}_{ij} = \Gamma^{\gamma}_{\alpha\beta}a^{\alpha}_{i}a^{\beta}_{j}\tilde{a}^{k}_{\gamma} + \frac{\partial a^{\gamma}_{i}}{\partial \tilde{u}^{j}}\tilde{a}^{k}_{\gamma}$$

也可以这样来看，由(1.1)式，两边与$\tilde{\mathbf{r}}_h$作内积，得$< \tilde{\mathbf{r}}_{ij}, \tilde{\mathbf{r}}_h >= \tilde{\Gamma}^{k}_{ij}\tilde{g}_{kh}$，即

(1.2) $$\tilde{\Gamma}^{k}_{ij} =< \tilde{\mathbf{r}}_{ij}, \tilde{\mathbf{r}}_h > \tilde{g}^{kh}$$

再利用$\tilde{g}^{kh} = g^{pq}\tilde{a}^{k}_{p}\tilde{a}^{h}_{q}$，代入验证即得。

3.平均曲率为$Weingarten$变化系数矩阵A的$trace/2$，而由55页(5.3)式知$A = (b_{\alpha\beta})(g^{\alpha\beta})$，故

$$H = \frac{1}{2}trA = \frac{1}{2}g^{\alpha\beta}b_{\alpha\beta}$$

4.$E = 1, F = 0, G = r^2$，曲面为正交参数，由78页(2.28)式得

$$\Gamma^{1}_{11} = \Gamma^{2}_{11} = \Gamma^{1}_{12} = \Gamma^{1}_{21} = \Gamma^{2}_{22} = 0$$
$$\Gamma^{2}_{12} = \Gamma^{2}_{21} = 1/r,\ \Gamma^{1}_{22} = -r$$

5.由

$$g_{ij} = \delta_{ij} + f_i f_j,\ \ g^{ij} = \frac{(1 + f_1^2 + f_2^2)\delta_{ij} - f_i f_j}{1 + f_1^2 + f_2^2}$$

所以Christoffel符号为

$$\begin{aligned}\Gamma_{ij}^{k} &= \frac{1}{2}g^{kh}(\partial_j g_{ih}+\partial_i g_{jh}-\partial_h g_{ij})\\ &= \frac{1}{2}\frac{(1+f_1^2+f_2^2)\delta_{kh}-f_k f_h}{1+f_1^2+f_2^2}(2f_h f_{ij})\\ &= \frac{(1+f_1^2+f_2^2)f_k f_{ij}-f_k f_{ij}f_h^2}{1+f_1^2+f_2^2}\\ &= \frac{f_k f_{ij}}{1+f_1^2+f_2^2}\end{aligned}$$

注意重复指标表示求和，即$f_h^2=f_1^2+f_2^2$

6.当(u,v)为正交曲率线网时，$F=M=0$，$Codazzi$方程化为

$$\begin{cases}(\frac{L}{\sqrt{E}})_v-N\frac{(\sqrt{E})_v}{G}=0\\(\frac{N}{\sqrt{G}})_u-L\frac{(\sqrt{G})_u}{E}=0\end{cases}$$

展开得

$$L_v=\frac{LG+NE}{2EG}E_v=HE_v,\ N_u=HG_u \tag{1.3}$$

7.平均曲率$H=\frac{\kappa_1+\kappa_2}{2}$为常数，若$\kappa_1=\kappa_2$为常数，则曲面为全脐曲面；
若$\kappa_1\neq\kappa_2$，不妨设$\kappa_1>H>\kappa_2$，取参数(u,v)为正交曲率线网，则有

$$F=M=0,\ L=\kappa_1 E,\ N=\kappa_2 G$$

则由第六题知

$$L_v=HE_v,\ N_u=HG_u$$

积分得

$$L=HE+f(u),\ N=HG-g(v)$$

即有

$$\begin{cases}(\kappa_1-H)E=f(u)>0\\(\kappa_1-H)G=g(v)>0\end{cases}$$

作参数变换

$$\begin{cases}\tilde{u}=\int\sqrt{f(u)}du\\\tilde{v}=\int\sqrt{g(v)}dv\end{cases}$$

则$r_u=r_{\tilde{u}}\sqrt{f},r_v=r_{\tilde{v}}\sqrt{g}$，所以$E=(\kappa_1-H)E\tilde{E}$，即$\tilde{E}=\frac{1}{\kappa_1-H}$，同理，$\tilde{G}=\frac{1}{\kappa_1-H}$，定义$\lambda=\frac{1}{\kappa_1-H}$，
则第一基本形式为

$$I=\lambda(d\tilde{u}d\tilde{u}+d\tilde{v}d\tilde{v})$$

又$L=<r_{uu},n>=<fr_{\tilde{u}\tilde{u}}+r_{\tilde{u}}\sqrt{f}',n>=f\tilde{L}$，即

$$\tilde{L}=\frac{L}{(\kappa_1-H)E}=\frac{\kappa_1}{\kappa_1-H}=1+\lambda H$$

同理

$$\tilde{N}=\lambda H-1$$

故第二基本形式为

$$II = (1+\lambda H)d\tilde{u}d\tilde{u} - (1-\lambda H)d\tilde{v}d\tilde{v}$$

8.在曲面上取正交曲率线网作为参数曲线网，则$I = Edu^2+Gdv^2, II = \kappa_1 Edu^2+\kappa_2 Gdv^2$，由Codazzi方程得到$E_v = 0, G_u = 0$，由此得到$K = 0$。若不是平面，则$\kappa_1, \kappa_2$中有一个不为零，作参数变换将两个基本形式写成最简单的形状，根据基本定理的唯一性部分证明是圆柱面。

9.验证φ和ψ是否满足$Gauss$方程和$Codazzi$方程，知没有这样的曲面；

10. $F(x,y,z) = 0$，局部可写成$z = z(x,y)$，对x求偏导得

$$\begin{aligned} F_x + F_z z_x &= 0 \\ F_{xx} + F_{xz}z_x + (F_{zx} + F_{zz}z_x)z_x + F_z z_{xx} &= 0 \end{aligned}$$

解出

$$\begin{aligned} z_x &= -\frac{F_x}{F_z} \\ z_{xx} &= \frac{-F_{xx}F_z^2 + 2F_{xz}F_xF_z - F_{zz}F_x^2}{F_z^3} = \frac{A}{F_z^3} \end{aligned}$$

同理

$$\begin{aligned} z_y &= -\frac{F_y}{F_z} \\ z_{xy} &= \frac{-F_{xy}F_z^2 + F_{xz}F_yF_z + F_{yz}F_xF_z - F_{zz}F_xF_y}{F_z^3} = \frac{B}{F_z^3} \\ z_{yy} &= \frac{-F_{yy}F_z^2 + 2F_{yz}F_yF_z - F_{zz}F_y^2}{F_z^3} = \frac{C}{F_z^3} \end{aligned}$$

于是

$$E=1+z_x^2, F=z_xz_y, G=1+z_y^2$$
$$L=\frac{z_{xx}}{\sqrt{1+z_x^2+z_y^2}}$$
$$M=\frac{z_{xy}}{\sqrt{1+z_x^2+z_y^2}}$$
$$N=\frac{z_{yy}}{\sqrt{1+z_x^2+z_y^2}}$$
$$K=\frac{LN-M^2}{EG-F^2}=\frac{z_{xx}z_{yy}-z_{xy}^2}{\left(1+z_x^2+z_y^2\right)^2}$$
$$K=\frac{AC-B^2}{F_z^2(F_x^2+F_y^2+F_z^2)^2}$$
$$K=-\frac{1}{\left(F_x^2+F_y^2+F_z^2\right)^2}\det\begin{bmatrix}F_{xx}&F_{xy}&F_{xz}&F_x\\F_{yx}&F_{yy}&F_{yz}&F_y\\F_{zx}&F_{zy}&F_{zz}&F_z\\F_x&F_y&F_z&0\end{bmatrix}=-\frac{1}{|\nabla F|^4}\det\begin{bmatrix}\nabla^2F&(\nabla F)^T\\\nabla F&0\end{bmatrix}$$

12.(1).曲面以φ和ψ为第一第二基本形式，则$F=M=0$，曲面以(u,v)为正交曲率线网，$L=\lambda E$，$N=\lambda G$，则平均曲率为$H=\lambda$，由第六题结论，$Codazzi$方程为

$$L_v=HE_v=\lambda E_v,\ N_u=HG_u=\lambda G_u \tag{1.4}$$

所以$\lambda_u=\lambda_v=0$，即λ是常数。

$Gauss$方程为

$$-\frac{1}{\sqrt{EG}}\{(\frac{(\sqrt{E})_v}{\sqrt{G}})_v+(\frac{(\sqrt{G})_u}{\sqrt{E}})_u\}=\lambda^2 \tag{1.5}$$

(2).当$E=G$时，$Gauss$方程化为

$$\Delta\ln E+2E\lambda^2=0 \tag{1.6}$$

由于$II=\lambda I$，曲面为全脐曲面，即为平面或球面。

当为平面时，$\lambda=0$，一个解为$E=G=1$；

当为球面时，$\lambda\neq0$，球半径为$\frac{1}{|\lambda|}$，一个解为$E=G=\frac{4}{(1+\lambda^2(u^2+v^2))^2}$（球极投影的解）。（这里通解无法写出，只能给出特解；题目叙述应修改）

13.

$$\begin{aligned}&e_1=\frac{1}{\sqrt{1+f_u^2}}(\cos v,\sin v,f_u),\ e_2=(-\sin v,\cos v,0)\\&e_3=\frac{1}{\sqrt{1+f_u^2}}(-f_u\cos v,-f_u\sin v,1)\\&\omega_1=\sqrt{1+f_u^2}du,\ \omega_2=udv\\&\omega_{12}=\frac{1}{\sqrt{1+f_u^2}}dv,\ \omega_{13}=\frac{f_{uu}}{1+f_u^2}du,\ \omega_{23}=\frac{f_u}{\sqrt{1+f_u^2}}dv\end{aligned} \tag{1.7}$$

14.$d\omega_{12} = -K\omega_1 \wedge \omega_2$，Gauss曲率与标架选取无关。

15.(1).直接计算知

$$\begin{aligned}
r_u &= (-a\sin u\cos v, -a\sin u\sin v, a\cos u)\\
r_v &= (-a\cos u\sin v, a\cos u\cos v, 0)\\
e_1 &= (-\sin u\cos v, -\sin u\sin v, \cos u)\\
e_2 &= (-\sin v, \cos v, 0)\\
e_3 &= e_1 \times e_2 = (-\cos u\cos v, -\cos u\sin v, -\sin u).
\end{aligned}$$

故(r, e_1, e_2, e_3)为一组正交活动标架。

(2).首先$\omega_1 = adu, \omega_2 = a\cos udv$. 而

$$\begin{aligned}
de_1 &= (-\cos u\cos v, -\cos u\sin v, -\sin u)du + (\sin u\sin v, -\sin u\cos v, 0)dv\\
de_2 &= (-\cos v, -\sin v, 0)dv,
\end{aligned}$$

于是

$$\begin{aligned}
\omega_{12} &= \langle de_1, e_2\rangle = -\sin udv\\
\omega_{13} &= \langle de_1, e_3\rangle = du\\
\omega_{23} &= \langle de_2, e_3\rangle = \cos udv.
\end{aligned}$$

(3).这时$\Pi = \omega_1\omega_{13} + \omega_2\omega_{23} = a(du^2 + \cos^2 udv^2)$.

16.在曲面S上取正交标架e_1, e_2是主方向，则有

$$\omega_{13} = \kappa_1\omega_1,\ \omega_{23} = \kappa_2\omega_2 \tag{1.8}$$

取$e_3 = n(u, v)$，由于曲面$\tilde{S}$与曲面S切平面平行，故不妨设$\{\tilde{e}_1, \tilde{e}_2, \tilde{e}_3\} = \{e_1, e_2, e_3\}$，则$\tilde{\omega}_{ij} = \omega_{ij}$。由

$$\begin{aligned}
d\tilde{r} &= dr + \lambda dn\\
&= \omega_1 e_1 + \omega_2 e_2 - \lambda\omega_{13}e_1 - \lambda\omega_{23}e_2\\
&= (1-\lambda\kappa_1)\omega_1 e_1 + (1-\lambda\kappa_2)\omega_2 e_2
\end{aligned}$$

故有$\tilde{\omega}_1 = (1-\lambda\kappa_1)\omega_1, \tilde{\omega}_2 = (1-\lambda\kappa_2)\omega_2$，所以$\tilde{\omega}_{13} = \dfrac{\kappa_1}{1-\lambda\kappa_1}\tilde{\omega}_1, \tilde{\omega}_{23} = \dfrac{\kappa_2}{1-\lambda\kappa_2}\tilde{\omega}_2$，即

$$\tilde{\kappa}_1 = \frac{\kappa_1}{1-\lambda\kappa_1},\ \tilde{\kappa}_2 = \frac{\kappa_2}{1-\lambda\kappa_2} \tag{1.9}$$

从而

$$\tilde{K} = \tilde{\kappa}_1\tilde{\kappa}_2 = \frac{K}{1-2\lambda H + \lambda^2 K},\ \tilde{H} = \frac{\tilde{\kappa}_1 + \tilde{\kappa}_2}{2} = \frac{H - \lambda K}{1-2\lambda H + \lambda^2 K} \tag{1.10}$$

18.$I = dudu + dvdv, \omega_1 = du, \omega_2 = dv$，所以$\omega_{12} = 0$，$Gauss$曲率$K = 0$

19.直接计算知

$$\omega_{12} = \langle de_1, e_2\rangle = -\frac{(\sqrt{E})_v}{\sqrt{G}}du + \frac{(\sqrt{G})_u}{\sqrt{E}}dv,$$
$$\omega_{13} = \langle de_1, e_3\rangle = \frac{L}{\sqrt{E}}du + \frac{M}{\sqrt{E}}dv,$$
$$\omega_{23} = \langle de_2, e_3\rangle = \frac{M}{\sqrt{G}}du + \frac{N}{\sqrt{G}}dv.$$

代入方程(6.6)即得Codazzi方程(3.17).

20.将$\omega_{13} = \kappa_1\omega_1$, $\omega_{23} = \kappa_2\omega_2$代入$Codazzi$方程即得。